Flight Dynamics of the Bell Boeing V-22 Osprey in JSBSim

A blade-element tiltrotor model, built and checked as the XV-15 from NASA's data and carried to the V-22 with its published rotor, wing and performance data

Technical report • October 2026

Abstract

A flight-dynamics model of the Bell Boeing V-22 Osprey for the FlightGear simulator has been written in JSBSim, replacing the parametric YASim model the aircraft used before. The V-22's engineering data are not public; those of its technology demonstrator, the Bell XV-15, are: Bell's real-time simulation data from its wind tunnel tests, NASA's hover test of its full-scale rotor and its flight tests. The model is therefore built and checked as an XV-15 first and carried to the V-22 with the V-22's published rotor geometry, wing, weights, engine ratings and performance charts. Each of the two three-bladed gimballed proprotors is simulated blade by blade, eight segments per blade in cambered-section tables at their own angle of attack and Mach number, under a Peters–He finite-state wake with ground effect; the gimbal's two degrees of freedom carry the pitch–flap coupling ($\delta_3 = -15°$). The nacelles tilt from 97.5° to airplane mode inside the conversion corridor; Bell's wing, fuselage and empennage tables, scaled, sit in the rotors' ring-vortex wakes; an interconnected drive with its torque limit, two engines, the collective governor and a stability augmentation complete it. The whole model, 614 aerodynamic functions and 550 control-system components, is generated by a script from transcribed data and runs in plain JSBSim at 480 Hz. As an XV-15 its isolated rotor needs −4.6 to +0.9 % of the measured hover power from $C_T/\sigma = 0.08$ to 0.18 (the section's stall fitted) and reaches $C_T/\sigma = 0.188$ against the test's 0.181; its wing's download follows the flight tests' within 1.0 point of thrust (fitted). In level flight, with the test altitudes unpublished, the model brackets the measured power through the conversion between sea level and 10 000 ft, but in airplane mode with the gear up it needs −35 to −24 % of it at 10 000 ft: Bell's airframe coefficients miss drag the aircraft had. As a V-22, with 11.5 ft^2 of parasite drag fitted to Boeing's airframe polar and the drive's torque limit fitted to the sea-level hover ceiling, its maximum level speed at 42 000 lb is 289 and 316 KTAS at sea level and 17 000 ft against 274 and 321, its hover ceiling at 8400 and 16 000 ft −2 and +5 % from Boeing's chart, and its pitch attitude through the conversion within 3.0° (rms) of the flight test's. Deficiencies are quantified: the XV-15's power in helicopter mode, +21 % at 45 kt and −29 % at 94 kt; its collective gradient in hover, 34 % steeper than measured; the V-22 wing's download, 11.7 % of thrust against 9.2; and the wing's share of the lift at 30° of nacelle.

1 Introduction

The V-22 distributed for FlightGear flies on YASim, whose helicopter model builds a rotor from a short list of geometric parameters and tuning factors and has no notion of a tiltrotor: the stock aircraft tilts its rotors' thrust and lets the wing take over. What makes the tiltrotor hard is precisely what such a model leaves out: two large, highly twisted proprotors that have to hover like a helicopter's rotor and cruise like propellers, the gimbal that lets each disc tilt as a whole, the wing that sits in their downwash in the hover (its download) and in their slipstream in cruise, and the conversion between the two, flown inside a corridor of nacelle angle against airspeed.

The V-22's own engineering data are not public. Its technology demonstrator's are: Bell documented the XV-15's real-time simulation in 1973 down to every wind tunnel table [1], NASA tested its full-scale rotor in hover [3], and NASA and the Army flew it and compared the simulation with the flights [4, 5]. For the V-22 itself Boeing and Bell published the rotor's geometry, the wing's planform, the airframe's lift and drag, the download, hover ceilings, the flight envelope [6] and the flight test results through the conversion [7]; NASA the rotor's characteristics [8, 9]. So the model is built as an XV-15 first, checked against the XV-15's tests, and then carried to the V-22 by its published data, each XV-15 quantity that has no V-22 counterpart scaled by geometry and marked as such.

This report describes the JSBSim [25] model built on that material, after UH-60A and Bo 105 models built the same way, and how well it reproduces the tests. All of the flight model, the blades' integration, the gimbals, the wakes, the engines and drive, the governor and the augmentation, is JSBSim XML; the FlightGear side only animates and instruments it. All result tables, figures and the numbers quoted from them are written by the evaluation scripts.

2 Reference data

Table 1 lists the sources; most are scans, and every curve used was digitised from them. A few points of data reduction matter for the comparisons. *(i)* Bell's simulation data [1] are complete but for one value: the wing–pylon pitching moment at zero lift for the third flap setting is illegible in the scan (page B-12) and taken as −0.20. *(ii)* The XV-15 rotor's hover test [3] gives its power as the report's curve fit through the wind-corrected data, C_P/σ against C_T/σ ($\sigma = 0.0891$), and its collective from the actuator's position, whose reference to a blade station is not given; it is compared here as a gradient. *(iii)* The XV-15's level-flight power [4] is published with neither the gross weight nor the altitudes; the flight envelope puts most of the programme between 2000 and 10 000 ft density altitude [5], so the model is flown at 13 000 lb at sea level and at 10 000 ft, which bracket it. Nor is the landing gear's position given; the gear is up throughout (Section 5.2). *(iv)* Dunford's conversion charts [7] give no weight, altitude or flap setting. Their airplane-mode pitch attitude, 13.6° at 108 kt falling to 0.2° at 170 kt, asks for a lift coefficient of 0.82 at zero angle of attack with Boeing's lift slope [6]: about 31 500 lb at sea level, below the MV-22B's empty weight. The comparisons run at 34 000 lb, sea level. *(v)* Boeing's hover ceiling and flight envelope [6] are for the 1986 design with 6150 shp engines and a standard day.

Table 1: Sources and what they contribute.

Source	Content used
NASA CR-114614 [1]	Bell's Model 301 (XV-15) simulation data: wing–pylon, fuselage and empennage tables, downwash, control mixing and rigging, governor, mass and inertia
NASA TM X-62407 [2]	XV-15 familiarisation: controls, flaperons, conversion range
NASA TM-86833 [3]	full-scale XV-15 rotor in hover: power against thrust, collective, blade twist, chord and sections
NASA TM-84222 [4]	XV-15 level-flight rotor shaft power from flight test by nacelle angle and flap (fig. 20)
12th ERF, paper 14 [6]	V-22 design data, rotor geometry and sections, wing, download against flap (XV-15 flight and V-22 0.658-scale test), airframe polar, hover ceiling, flight envelope
16th ERF, II.7.3 [7]	V-22 control system, conversion corridor, rotor speeds, airframe lift share and pitch attitude through the conversion, power required
NASA/TM-2016-219070 [8]	JVX (0.656-scale V-22) rotor characteristics, precone
Johnson, AHS 2001 [9]	1/4-scale V-22 rotor (TRAM): sections, δ_3, chord and twist
V-22 fact sheet [10]; GlobalSecurity [11]	weights, fuel capacity, crew; the MV-22's tanks and their transfer sequence
NACA R-824 [12]	64-series sections' lift, stall and drag (the sections' parameters)
NASA TN D-234, NACA TR 1319 [13, 14]	ground effect in forward flight; ring-vortex wake
Peters and He [15], via [16]	finite-state dynamic wake

3 The model

3.1 Implementation

A Python script writes the model from the transcribed tables (Table 2), the XV-15 or the V-22 from the same code with each aircraft's data. JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past, the gimbals' tilt and rate, the rotor azimuth and speed, the wake states, the nacelle angle, the engines' states, the governor's and the augmentation's integrators, therefore lives in flight-control-system channels, which JSBSim runs once per frame in a fixed order and which can write properties back. The blade elements, the gimbals and the drive train integrate explicitly at 480 Hz, the rate FlightGear runs the model at: at 104 % rotor speed 5.2° of azimuth per step. The rest runs every few frames, which JSBSim's channels allow: the wake states with their gains and forcing (time constants of a tenth of a second and more) and the controls, mixing, servos, governor and augmentation, every fourth (120 Hz), the airframe with its wake and ground-effect tables, the engines and the outputs to FlightGear every eighth. In FlightGear every property read takes a lock, so a step costs what it reads; with these rates and the blade elements' terms arranged for it (the first wake shape's and the flap angle's share of the normal velocity taken per blade, the angle of attack into tables carried on to ±360° instead of wrapped) a step reads about 3200 properties, 4300 at the FDM rate throughout; the ceilings, speeds and download are the same to five digits, the trims within 0.2 % of power and 0.04° of attitude. The same XML runs in FlightGear and in the offline harness used here; FlightGear sets only the pilot's controls and reads the outputs.

Table 2: The generated V-22 model: JSBSim system files, their content and the number of named aerodynamic functions or control-system components.

File	Content	n
`-aero.xml`	the aerodynamic functions: blade segments of both rotors, gimbal moments, wing strips in the wakes, fuselage, empennage; the six axes	614
`-rotors.xml`	gimbals, rotor azimuth, Peters–He wake states of each rotor with their gains and forcing	77
`-airframe.xml`	ring-vortex wake at the wing, stabiliser and fins; ground effect	157
`-drive.xml`	engines, torque limit, interconnected drive and rotor speed, rotor brake, fuel	37
`-controls.xml`	pilot controls, mixing, nacelles and conversion protection, flaperons, governor, augmentation, hover hold, servos	237
`-outputs.xml`	properties for the 3-D model and the instruments	42

3.2 Proprotors

Each rotor's three blades are rigid and sit with their precone (2.5° [8, 3]) on a gimbal, so the disc tilts as a whole, as on both aircraft. The gimbal's two degrees of freedom, in the non-rotating frame, are integrated from the blades' aerodynamic flapping moments about the hub, the gyroscopic coupling of the shaft's rotation (taken exactly, as a rotation of the tilt each step), the airframe's angular rates and a hub spring; each blade's pitch takes $\tan\delta_3$ of its own flapping, $\delta_3 = -15°$ [1, 9]. The blades' inertia about the gimbal is the XV-15's Lock number's (3.8), scaled. Eight segments of equal annulus area from 0.15 R (V-22; 0.1 R XV-15) see the air velocity at their midpoint from the full kinematics: the aircraft's motion, the nacelle's tilting rate, the rotor's rotation, the gimbal's tilt and rate and the induced velocity. The left rotor is the right one's mirror image and is computed as the right rotor of the mirror-image aircraft.

Neither rotor's section tables are public: the XV-15's metal blades use NACA 64-series sections [3], the V-22's Boeing's XN sections of 28 % thickness at 0.2 R to 9 % at the tip [6, 9]. Each segment's section is built as C-81's equations build the NACA 0012, with the parameters of a cambered section of its thickness and design lift [12]: a lift slope and zero-lift angle from the camber, a maximum lift falling with thickness and Mach number, a trailing-edge stall that rounds over into the maximum and falls gently past it to a flat plate's normal force, Korn's drag divergence. The stall's shape, its maximum lift ($C_{l\max} = 1.55$ at zero camber) and the separated drag are *fitted* to the XV-15 rotor's hover power at high thrust (Section 5.1). The V-22's twist, 47.5° from the root to the tip, and its chord, 34.3 in extrapolated to the centre to 22 in at the tip, are Boeing's [6, 9]; the tip loss factor is 0.97.

3.3 Inflow and wake

Each rotor has its own Peters–He finite-state wake [15, 16], three axisymmetric states and the first harmonic pair, in the shaft's frame as the nacelle tilts. Ground effect scales the induced velocity by $(1+0.3(R/Z)^2)^{-1/2}$ in the hover, with Z the hub's height, carried into forward flight with Heyson's interference ratios [13]. Each rotor's downwash at the wing, the horizontal stabiliser and the two fins comes from a skewed cylinder of ring vortices leaving the disc [14], contracting below it in the hover: averaged over four spanwise strips of each wing half, the stabiliser and each fin, and tabulated against the wake's skew, its direction in the disc's plane and the nacelle angle, which moves the hub and turns the disc.

3.4 Airframe

The wing–pylon, fuselage, horizontal stabiliser and fin data are Bell's tables for the XV-15 [1]: the wing's lift and drag against angle of attack, flap setting and mast angle, its downwash at the tail, the fuselage's forces and moments in equation form, the stabiliser's and fins' coefficients with their blockage. For the V-22 the wing's are applied to its own area, span and chord (382 ft^2, 45.8 ft [6]), the empennage's to its surfaces, and the fuselage's scaled by the length ratio squared and cubed. Each wing half is four strips in the rotors' wakes, with the local flow's angle of attack. In the hover the wakes push the wing down; the download is Bell's flap-dependent coefficient in the wake's velocity times a factor *fitted* to the XV-15's download in flight for each flap setting (Section 5.3). Bell's coefficients miss part of the V-22's parasite drag: 11.5 ft^2 is added, *fitted* to Boeing's airframe polar, flaps and gear up (Section 5.4). The landing gear adds Bell's drag areas, scaled, when extended.

3.5 Flight controls

The cockpit controls' mixing is Bell's for the XV-15 [1], in its control travel: the thrust control lever sets both rotors' collective (its slope falling to zero in airplane mode), the lateral stick their differential collective, the longitudinal stick their longitudinal cyclic, the pedals their differential cyclic, each phased with the mast angle, while the elevator, flaperons and rudders follow the stick and pedals in all modes. The nacelles move at up to 8°/s from 97.5° (7.5° aft of vertical) to 0, their command held within the conversion corridor against equivalent airspeed [7]. The flaperons' schedule is not public; they are at 67° in the hover for the least download [6], at 40° through the conversion and in airplane mode below 120 kt (the corridor's lower boundary is the wing's stall at 40° of flap [7]) and retract by 200 kt (an estimate). A collective governor holds the rotor speed's reference, 104 % in helicopter and conversion modes and 84 % in airplane mode [7], proportional and integral within Bell's collective limits.

The V-22 flies by wire and holds attitude in all modes [7]; its control laws are not public. The FlightGear aircraft has a stability augmentation of a common rotorcraft structure: rate damping in three axes for a 0.5 s rate time constant in the hover; attitude command and heading hold below 50 kt with the nacelles above 60° and rate command with attitude hold above (switching at 55 kt accelerating and 45 kt slowing, so that a speed held near 50 does not switch back and forth), switched without a transient because both share one reference [22]; in yaw below 50 kt rate command, direction hold, the pedals commanding a yaw rate (30°/s at full travel) held against the wind by an integrator, the hover's response type in ADS-33E [17]; a feed-forward of the level-flight trim's stick against nacelle angle and forward airspeed, from the model's own trims, so that the stick's centre is trimmed flight through the conversion; and integrators that take up the rest. A hover hold (Section 5.8) and an autopilot (Section 5.9) work through it. The comparisons of Section 5 fly the bare airframe, the augmentation off.

Table 3: Principal data of the two builds.

Item	V-22	XV-15
Rotor radius, ft	19.04	12.50
Blades per rotor	3	3
Precone, deg	2.50	2.50
δ_3, deg	-15	-15
Chord root (extrapolated) / tip, in	34.3 / 22.0	14.0 / 14.0
Rotor speed 100 %, rpm	397	589
Rotor speed, helicopter/airplane, %	104 / 84	100 / 88
Wing area, ft^2; span, ft	382; 45.83	181; 32.17
Engines, shp each	6150	1550
Drive torque limit at 100 %, hp	9130 (fitted)	2500
Nacelle angle range, deg; rate, deg/s	0–97.5; 8	0–95.0; 8
Reference gross weight, lb	47500	13000
Empty weight, lb	33140	–
Fuel, US gal (tanks)	1448 (5)	–

3.6 Engines, drive and mass

The engines are not modelled in detail: the lever sets each one's power demand from flight idle to its rating (6150 shp each [6]), falling with altitude as $\delta/\sqrt{\theta}$; the engine follows with a 0.6 s lag, and a topping governor takes power off above 2 % of overspeed. The interconnected drive gears both rotors to one speed, with the rotors', the power turbines' and the interconnect shaft's inertia; the drive's torque limit, shared between the running engines, is *fitted* to Boeing's sea-level hover ceiling (9130 hp at 100 % rotor speed; Section 5.5). A start brings each engine from rest to flight idle in 30 s. The MV-22B's five tanks (1448 US gal [10, 11]) are drained in the transfer sequence's order. Each nacelle, 7000 lb with its engine, gearbox and rotor (15 % of the gross weight, as the XV-15's pylons [1]), tilts with its centre of gravity and inertia. The reference loading is 47 500 lb, the 1986 design's maximum vertical take-off weight [6], at a centre of gravity under the hubs in helicopter mode: the empty weight (33 140 lb [10]), three crew, the tanks full but for the right aft sponson and the rest as cargo. The inertias are the XV-15's scaled by mass and dimensions. Table 3 lists the principal data.

4 Method

The evaluation drives JSBSim from Python with the same files as FlightGear, at 480 Hz, the augmentation off and the gear up (in the hover its drag is nil). A trim solves for the collective, the cyclic or elevator, the pedals, the angle of attack and the roll that zero the six body accelerations, averaged

Table 4: The isolated XV-15 rotor in hover: power and figure of merit at the model's thrust.

θ_{75}	C_T/σ	C_P/σ, model	test	Δ %	FM, model	test
8.0	0.085	0.00704	0.00698	+0.9	0.742	0.748
9.0	0.097	0.00828	0.00831	−0.3	0.769	0.767
10.0	0.109	0.00967	0.00971	−0.4	0.786	0.783
11.0	0.121	0.01122	0.01138	−1.4	0.796	0.784
12.0	0.134	0.01291	0.01324	−2.5	0.799	0.779
13.0	0.146	0.01476	0.01527	−3.3	0.795	0.769
14.0	0.157	0.01679	0.01761	−4.6	0.783	0.746
15.0	0.167	0.01900	0.01982	−4.2	0.761	0.729
16.0	0.176	0.02147	0.02234	−3.9	0.726	0.698

over whole rotor revolutions (Hann-weighted), with the airframe held and the rotor at the governed speed, by Newton's method with a backtracking line search. The programme:

- *the XV-15 rotor in hover*: the XV-15 build held far from the ground at the test's tip Mach number (589 rpm, 1000 ft), swept in collective; the right rotor's thrust and power;
- *XV-15 level-flight power*: trims at 13 000 lb at sea level and 10 000 ft by calibrated airspeed for each of the test's nacelle and flap settings, the flaperons at 0.625 of the flap [2];
- *download*: hover trims out of ground effect at sea level, the XV-15 at its flight tests' flap settings, the V-22 at 67°;
- *V-22 airframe polar*: the airframe's own forces in airplane mode at 198 KTAS (Mach 0.3) and 3000 ft, flaps and gear up, the rotors at zero thrust;
- *V-22 performance*: the maximum level speed in airplane mode at 84 % rotor speed, where the power required meets the engines' power or the drive's limit; the hover ceiling out of ground effect at 100 % rotor speed (the 1986 chart's); trims through the conversion at 34 000 lb by nacelle angle;
- *handling* with the augmentation on: single keyboard steps in the hover, a hands-off conversion, an engine start;
- *the hover hold*: its stops from up to 39 kt, ADS-33E's position hold turn in up to 35 kt of wind, its stick and lever;
- *the autopilot*: coupled from trims at 1000 ft, an altitude change, a turn, the conversion down and up under AUTO NAC; approaches to a hover and landings, a route's, go-arounds; a route's turns; terrain following over profiles of terrain set under the aircraft as it flies, with the plan of the FlightGear script mirrored (every 0.25 s, and with the timer up to a frame late).

5 Results

5.1 The XV-15 rotor in hover

From $C_T/\sigma = 0.08$ to 0.18 the rotor needs −4.6 to +0.9 % of the test's power at the same thrust (Fig. 1, Table 4), the figure of merit peaking at 0.80 at $C_T/\sigma = 0.13$ against the test's 0.79 at 0.12, and the rotor reaches $C_T/\sigma = 0.188$ at 19° of collective, where the test, which stopped at 0.181, did not go. The sections' stall was fitted here, so this checks the shape of the curve rather than predicts it. Below $C_T/\sigma = 0.075$ the model needs 3–21 % more power than the test, more the lower the thrust: the profile power of the inboard sections, whose thick XN-like stand-ins have more drag at low lift than the 64-series. The thrust's gradient against collective is 0.0114 per degree against the test's 0.0085 (34 % steeper): the test's collective is the actuator's, which includes the blades' elastic twist under load, which the rigid blades do not have. Past the maximum thrust the blades stall progressively and the gimbal whirls by a few degrees; the model's power there (the points that turn back in Fig. 1a) has no test to compare with.

Table 5: XV-15 level-flight power, model/test −1 at the test's points.

Nacelle, flaps	KCAS	sea level, %	10 000 ft, %
90°, flaps 20°	0–94	−29 to +21	+7 to +42
60°, flaps 20°	97–134	−19 to −7	+2 to +9
30°, flaps 20°	126–167	−13 to −8	+1 to +8
0°, flaps 20°	134–180	−23 to −14	−14 to −7
0°, flaps up	156–232	−41 to −29	−35 to −24
0°, flaps up, gear down	156–232	−20 to −6	−13 to +3

Table 6: Download in the hover, % of the rotors' thrust.

Aircraft	flap, deg	model, %	test, %	Δ
XV-15	0	16.3	16.4	−0.1
XV-15	40	13.0	13.0	−0.0
XV-15	60	11.4	10.4	+1.0
XV-15	75	10.2	10.5	−0.3
V-22	67	11.7	9.2	+2.5

5.2 XV-15 level-flight power

Through the conversion, nacelles at 60° and 30° with 20° of flap, the test lies between the model at sea level and at 10 000 ft (Fig. 2, Table 5): −19 to −7 % and −13 to −8 % at sea level, +2 to +9 % and +1 to +8 % at 10 000 ft, as the test's altitudes would have it. In airplane mode the model needs less power than the test at either altitude: −23 to −14 % with 20° of flap at sea level and, flaps up, −41 to −29 % at sea level and −35 to −24 % at 10 000 ft. With the gear extended, its 4 ft^2 of drag [1] brings the flaps-up power to −13 to +3 % at 10 000 ft; the test's gear position is not given, but in airplane mode it would have been up. Bell's airframe coefficients, from wind tunnel tests of the clean model, miss part of the flying aircraft's drag; the V-22's needed 11.5 ft^2 more to meet Boeing's polar for the same reason (Section 5.4). The XV-15 build keeps Bell's data as they are.

In helicopter mode the model's power falls too fast with speed: +21 % at 45 kt, −29 % at 94 kt at sea level, +7 % in the hover. Bell's own simulation used an empirical profile power for the rotor in this regime; the blade elements here give a translational lift that lowers the induced power faster than the test, and the wing in the slipstream at a high angle of attack, with 20° of flap, sees the wake model's velocities, whose sweep off the wing through the transition is not checked by any test.

5.3 Download

The download factor of each of Bell's flap settings was fitted to the XV-15's flight-measured download, which the model then follows within 1.0 point of thrust (Fig. 3a, Table 6). Carried to the V-22 with its own wing and rotors, at the hover's 67° of flaperon, it gives 11.7 % against the 0.658-scale test's 9.2 %: the V-22's flaperons, which fold down over a larger part of the chord than the XV-15's flaps, take more of the

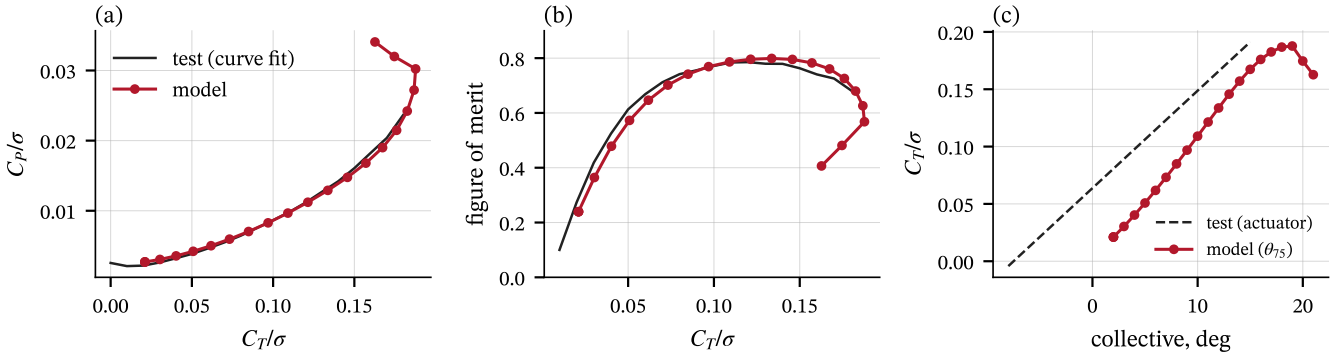


Figure 1: The isolated XV-15 rotor in hover at a tip Mach number of 0.69 against NASA's full-scale test (TM-86833 figs. 13, 14, 17): (a) power against thrust, (b) figure of merit, (c) thrust against collective (the test's from the actuator's position, the model's at 0.75 R).

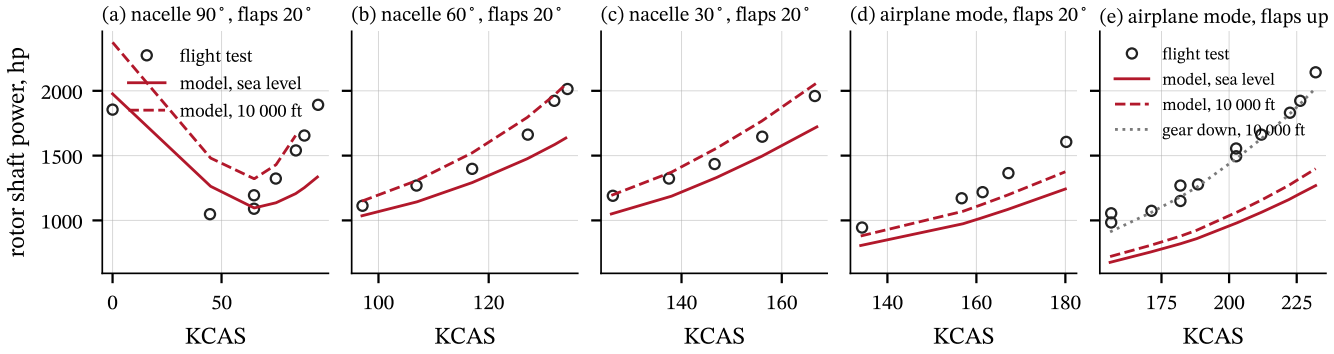


Figure 2: XV-15 rotor shaft power in level flight against the flight test (TM-84222 fig. 20) at 13 000 lb: the model at sea level and at 10 000 ft, which bracket the test's unpublished altitudes, gear up; in (e) also with the gear down.

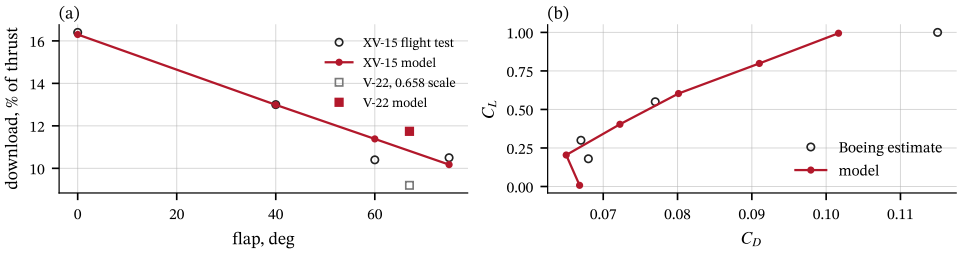


Figure 3: (a) The wing's download in the hover against flap angle: the XV-15 (13 300 lb, flight test [6] fig. 18) and the V-22 at 67° (0.658-scale rotor and wing test). (b) The V-22 airframe's lift and drag in airplane mode, flaps and gear up, against Boeing's full-scale estimate [6] fig. 28.

Table 7: V-22 airframe drag coefficient at Boeing's lift coefficients, on 382 ft^2; Δ in drag counts.

C_L	C_D model	C_D test	Δ, counts
0.18	0.0652	0.0680	−28
0.30	0.0685	0.0670	+15
0.55	0.0780	0.0770	+10
1.00	–	0.1150	–

wing out of the wake than the XV-15's tables know.

5.4 V-22 airframe polar

Bell's wing and empennage tables applied to the V-22's surfaces give it a lift coefficient of 0.21 at zero angle of attack and a slope of 0.099 per degree, against Boeing's 0.18 and about 0.093 (Fig. 3b, Table 7). Its drag, with the 11.5 ft^2 fitted here, is 0.0651 at $C_L = 0.2$ against 0.068; the model's polar is flatter than Boeing's estimate, whose drag rises faster beyond $C_L = 0.6$.

Table 8: V-22 performance, model and Boeing's charts.

Quantity	model	test [6]
maximum level speed, 42000 lb, 0 ft	289 KTAS	274 KTAS
maximum level speed, 47500 lb, 0 ft	286 KTAS	–
maximum level speed, 42000 lb, 17 000 ft	316 KTAS	321 KTAS
hover ceiling OGE at 0 ft	47690 lb	47 700 lb
hover ceiling OGE at 4 000 ft	45411 lb	–
hover ceiling OGE at 8 400 ft	42286 lb	43 300 lb
hover ceiling OGE at 12 000 ft	37623 lb	–
hover ceiling OGE at 16 000 ft	32579 lb	31 000 lb

5.5 V-22 performance

The drive's torque limit was fitted to the hover ceiling at sea level, 47 700 lb, where Boeing's chart has its knee: the transmission limits there, the engines above. With it the ceiling is 42 290.0 lb at 8400 ft (−2 %) and 32 580.0 lb at 16 000 ft (+5 %), where the engines' lapse sets it (Fig. 4, Table 8). At 42 000 lb the maximum level speed is 289 KTAS at sea level against 274, limited by the drive's torque at 84 % rotor speed,

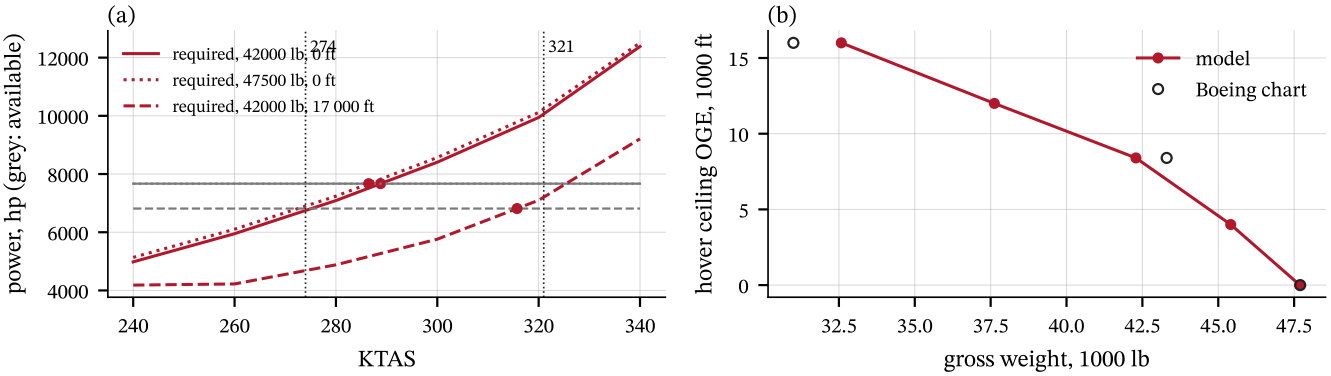


Figure 4: V-22 performance against Boeing's charts [6] (standard day, 6150 shp engines): (a) power required in airplane mode at 84 % rotor speed and power available (grey: the engines' rating or the drive's torque limit, whichever is less); the dotted verticals are the chart's maximum speeds at 42 000 lb. (b) Hover ceiling out of ground effect at 100 % rotor speed.

Table 9: Conversion, model minus test.

Nacelle, deg	KCAS	wing lift, % of W	pitch, deg
90	0–82	−11 to −1	−3.2 to +4.5
75	37–101	−11 to +1	−2.8 to +3.4
60	60–126	−7 to −0	+0.1 to +3.9
30	110–168	−25 to −10	+3.0 to +6.1
0	116–184	−10 to −4	+0.6 to +4.7

and 316 KTAS at 17 000 ft against 321, limited by the engines.

5.6 The conversion

The pitch attitude follows the flight test through the conversion within 3.0° (rms; Fig. 5, Table 9), the model a few degrees nose-up of it at the higher speeds of each nacelle angle. The wing carries 8 % of the weight less than Dunford's chart (rms), most at 30° of nacelle, where the model's wing carries 62–72 % against 83–87 %: at that angle the rotors' slipstream over the wing, which the wake model gives, decides the wing's lift, and no test checks it. The charts' weight and flaps are not published (Section 2).

5.7 Handling with the augmentation, and the start

These are not comparisons but checks of the FlightGear aircraft's augmentation (Fig. 6). In the hover one press of the stick forward gives −1.1° of pitch, held, and one to the right 1.4° of roll after a small overshoot; one of the pedals turns it at 1.6°/s and the heading holds where they are let go. With the nacelles commanded from 90 to 60° at 40 kt and nothing else touched, the aircraft accelerates to 158 kt with its pitch between −7.2 and 1.9°: the trim feed-forward takes up the trim's change, the stick's trim moving from 65 % of its travel aft of centre in the hover at 75° to 76 % forward at 120 kt at 90°, which the bare aircraft would have the pilot hold. The nacelles stop at the corridor's lower boundary until the airspeed allows 60°, and later move on past it as the upper boundary comes down. Cold, with the lever down, the engines spool up one after the other and the rotors reach their governed speed 74 s after the first engine's start.

Table 10: Terrain following, 200 ft set, the medium ride; after the first 5 s. Jitter: the least clearance with FlightGear's timer up to a frame (77 ms) late.

Terrain	KCAS	clearance, ft: least, mean, most			jitter	vertical g	pitch, deg
level	120	200	200	200	200	−0.01 to +0.01	+2.8 to +2.9
level	150	199	200	201	–	−0.01 to +0.01	+3.7 to +3.9
level	200	200	200	200	200	−0.01 to +0.01	+6.1 to +6.2
hills	120	198	200	202	–	−0.12 to +0.10	+0.5 to +5.5
hills	150	193	231	341	–	−0.26 to +0.23	−3.8 to +9.3
hills	200	195	325	500	–	−0.20 to +0.11	+0.7 to +13.0
ridge	120	196	254	620	195	−0.23 to +0.20	−1.1 to +6.8
ridge	150	195	277	722	–	−0.23 to +0.20	−4.0 to +9.4
ridge	200	194	292	976	194	−0.20 to +0.15	+0.3 to +13.4
ridge	170	178	277	885	–	−0.31 to +0.11	−1.5 to +14.3

5.8 The hover hold

The hover hold (HVR CPLD) follows the AH-64D's modern control laws [18], Bell's automatic hover hold [20] and ADS-33E's position hold and translational rate command [17], with the V-22's own way of translating: on its nacelles, the fuselage level [21], as NASA's large tiltrotor's translational rate command does [19]. Engaged at up to 40 kt over the ground, it sets the stop point, where the present velocity comes to rest at 0.08 g, and flies one profile to it, then holds it and the height (radar height below 500 ft, the barometric altitude above); the nacelles take the fore and aft acceleration, the bank the sideways, the pitch only what is past the nacelles' range. Engaged at 39 kt it stops 867 ft on, holding 26 s after; every stop ends within 0.6 ft of the point it showed, the pitch within 2.4° (Fig. 7a). In ADS-33E's check, a turn on the pedals in about 30 s, the aircraft stays within 4.5 ft of the point in still air, which meets the 10 ft circle, 6.7 ft in 20 kt of wind and 10.1 ft in 35 kt, about twice the circle's radius, its attitudes within 6.6° against the 18° allowed (Fig. 7b). The stick out of its detent commands a ground speed: 5.1 kt for a quarter of its travel, reached to 63 % in 3.8 s (2.5–5 s), overshooting by 0.2 kt, and on its release the aircraft stops and holds where it does (Fig. 7c). The hold drives the lever and the pilot can fly through it: moved for 4 s and left, the aircraft settles 7 ft lower, held within 0.1 ft (Fig. 7d). The hold's sideways loop works through the roll attitude's, which needed twice its first gains; the 35 kt turn's errors come from the wind's loads turning with the aircraft faster than the hold's integrators follow.

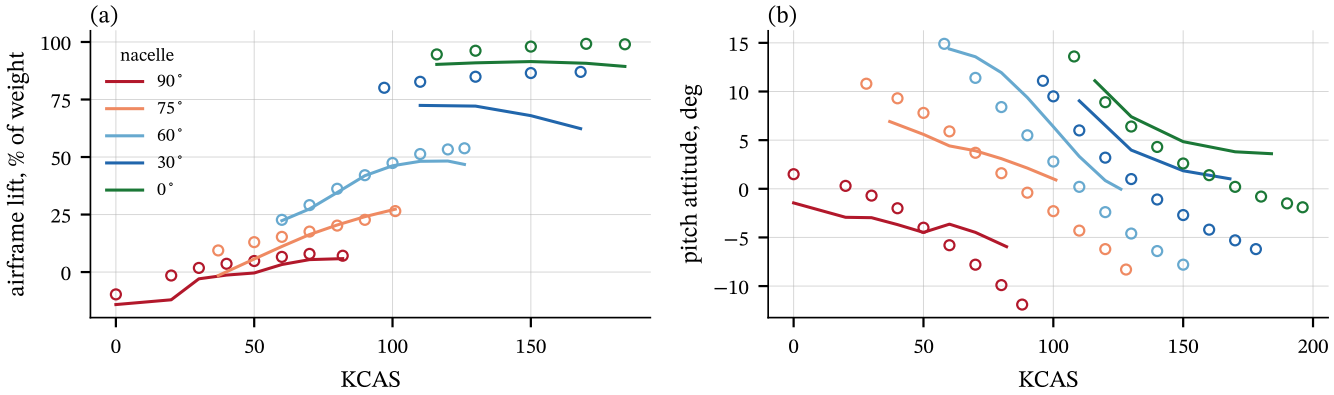


Figure 5: Trims through the conversion at 34 000 lb, sea level, against the V-22's flight test [7] figs. 7 and 9a: (a) the airframe's share of the lift, (b) pitch attitude, by nacelle angle (circles: test).

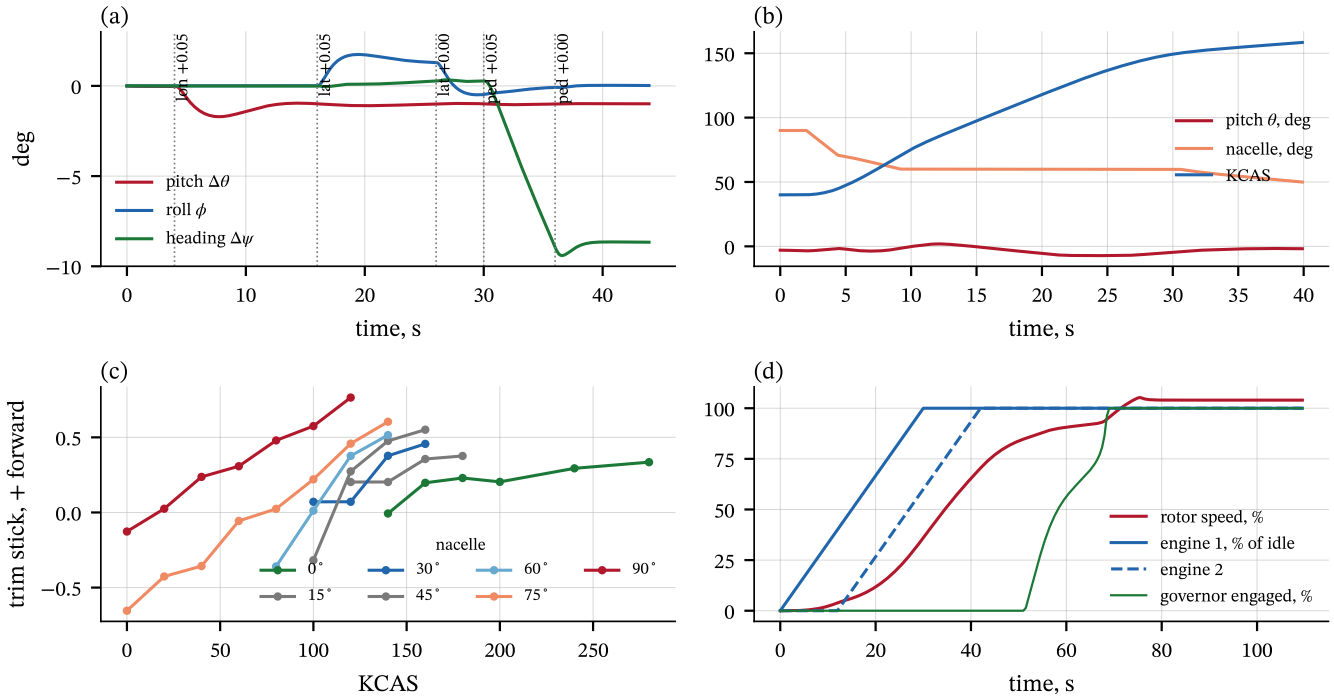


Figure 6: The FlightGear aircraft, augmentation on, reference loading: (a) single keyboard steps (5 % of travel) from the hover, stick centred; (b) a hands-off conversion from 40 kt, the nacelles commanded from 90 to 60°; (c) the trim feed-forward, the level-flight trim's stick against airspeed by nacelle angle; (d) an engine start on the ground, lever down.

5.9 The autopilot

The autopilot is a flight director that the augmentation flies when coupled, as the V-22's is [7]; its laws are not the V-22's, which are not public. In airplane mode the lever and the pitch share the speed and the flight path by total energy control [24]: the lever the total energy's rate, the pitch its share between climb and speed, the commanded flight path's angle fed forward. Blended by the nacelle angle from 50 to 80° it passes to the back side of the power curve, where the lever flies the climb and the pitch the speed. AUTO NAC takes the nacelle angle for the speed from the model's own level trims, the most converted angle whose trim's pitch is at most 4° (7° at 180 kt), within the corridor. The autopilot commands the augmentation's attitudes and their rates; the stick for those rates is fed forward from the airframe's measured response per inch of stick, and the augmentation damps the difference from the commanded rate rather than the rate itself, which had resisted every turn and pull-up. In pitch the airframe's first response to the stick is up to seven times its sustained rate (at 150 kt, 50°); the feed-forward goes through a lag and closes its error over a time both set from that ratio, which keeps the loop's gain at high frequency below one: fed straight it rang at 1 Hz and diverged at 150 kt.

Coupled at 200 kt it climbs 1000 ft without overshoot, its pitch at most 10.2°, and turns 90° banked at most 27.1° within 11 ft of its altitude. Converting from 220 to 100 kt and back under AUTO NAC it holds the altitude within 50 ft, the pitch between −2.4 and 7.6°. Engaging, the vertical speed's and the speed's references start from the aircraft's own vertical speed and acceleration and change at no more than 0.1 g, the speed's in the energy's terms, $V\dot{V}/g$ (its rate changing at no more than 0.1 g^2/V, easing onto the selected speed too): stepped, the speed's rate to 2 kt/s at once and back to 0 on reaching it, an engagement at 120 kt climbing at 32 ft/s and 8 kt slow, as after FlightGear's in-air start, pitched the nose from +7.6 to −3.8° and back to +3.6°, ringing on. Now it pitches from +7.6 to −1.6° and comes back by at most 0.8°, within 0.14 g. An approach to a point along a course and a glide path (an ILS's, or a runway's or landing zone's, from 6 nm and 2000 ft

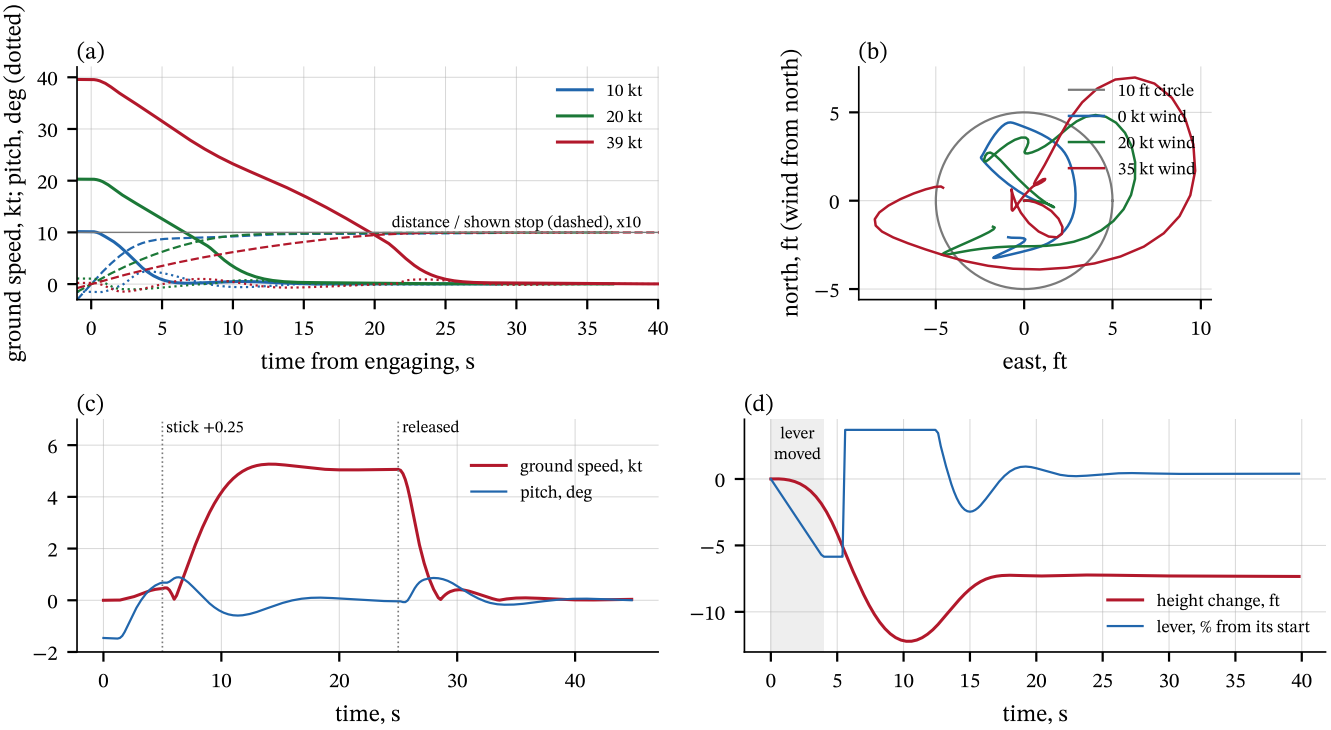


Figure 7: The hover hold, reference loading, augmentation on: (a) engaged from 10, 20 and 39 kt, the ground speed, the pitch attitude (dotted) and the distance flown over the stop it showed (dashed); (b) ADS-33E-PRF's position hold check, a 360° turn on the pedals in about 30 s in a steady wind from the north, the track about the hold point; (c) the stick out of its detent and back; (d) the lever moved down for 4 s and left.

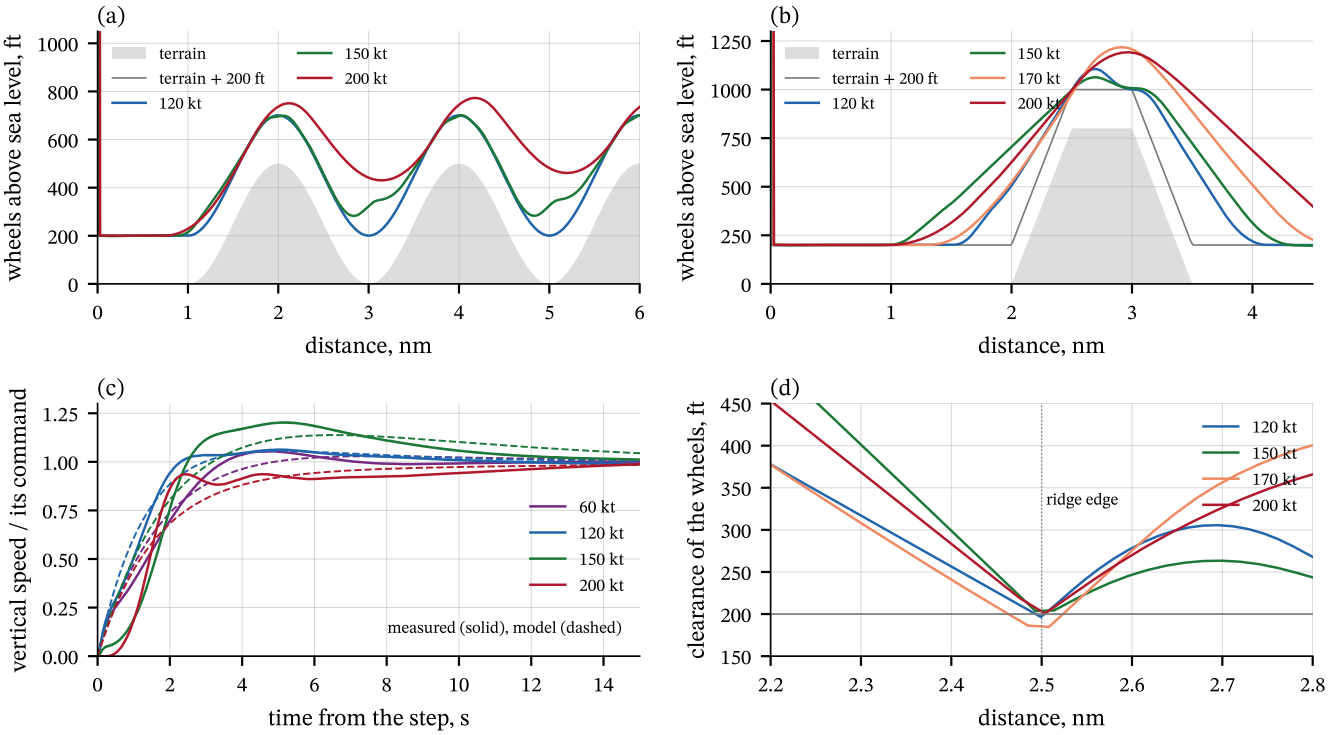


Figure 8: Terrain following at 200 ft, the medium ride (0.2 *g*), SPD HOLD and AUTO NAC, reference loading: (a) over hills 500 ft high and 2 nm apart; (b) over a ridge 800 ft high with 15° faces; (c) the climb's response to a step in TF's command (10 ft/s), measured, and the model the flight model keeps of it; (d) the clearance at the ridge's edge.

off the line) keeps within 8 ft of the course and −29 to 19 ft of the path, slows down along it and hands over to the hover hold below 35 kt over the ground (35 kt of airspeed in still air, 49 kt in 15 kt of headwind), which holds the hover within 5 ft of the point or lands 0.7 ft from it, sinking at 1.7 ft/s. A go-around loses at most 3 ft. Flying a route, the speed comes down from 1.16 nm before its landing zone, where the plan drawn on the navigation display has its deceleration point at 1.33 nm, and the aircraft lands 0.2 ft from the point. Between legs it flies its own arcs, started where an arc at 80 % of the bank limit joins them, led by the roll's measured delay: past the next leg by at most 74 ft at 200 kt and 22 ft at 100 kt in 90° turns, and

36 ft in a 149° turn at 120 kt.

Terrain following (TF) holds a set clearance over the scenery's terrain. Every 0.25 s it samples the terrain along the predicted track (curving at the present turn rate) every 1.25 s of flight to 45 s ahead, on the track and 150 ft either side, and keeps the last four sweeps' points, which lie between them 0.25 s apart: one sweep's alone missed a ridge's edge by up to 1.25 s. It does not plan from the aircraft. Planned from the aircraft's state, terrain nearer than its climb's delay could be met only with everything or nothing, a relay through a delay, and the aircraft porpoised over level ground, ±8 ft/s at 200 kt. The flight model instead keeps a reference path, its vertical speed ramped at the ride's acceleration to the plan's command, and a model of the aircraft following it: two first-order lags in parallel fitted to the climb's measured response to a step, the pitch's quick share and the slower part the lever adds (Fig. 8c, rms within 0.08). The aircraft is held to the model's height with a gain set for 60° of phase margin on that measured response, 0.28–0.63/s. For a vertical speed commanded now, reached at the ride's acceleration and held to the next plan, the plan has the model's height at each point in closed form, with the most a pull-up at half the ride's acceleration then adds up to the climb the power makes, and commands the least vertical speed that clears every point. It descends into a valley and pulls up as late as half the ride allows. A point too near to clear in time, nearer than the height loop's time constant, is a height error, closed over that time with the command not eased meanwhile: FlightGear's timer, up to a frame late, otherwise left the model a fraction of a foot under the clearance and the plan climbing at full power for it. Where the lever reaches its stop and the aircraft still falls behind, the model and the reference come down to it, and the plan never counts on height the aircraft has not got. The climb the power makes comes from the power a climb takes, $P = P_{\text{level}} + b\dot{h} + c\dot{h}^2$ (b, c from three trims at each speed, the level flight's power from the present power and the energy's rate), divided by the share of a climb at constant calibrated airspeed that goes into the true airspeed's rise (5 % at 200 kt). These tables are read at the speed AUTO NAC's schedule gives the present nacelle angle: through the conversion the nacelles change the climb more than a few knots do, 2058 and 2734 ft/min at full power at 160 and 170 kt. With SPD HOLD the speed is free within 20 kt of its reference. Below half the clearance, or without terrain data, it climbs at full power.

Table 10 and Fig. 8 give TF over level ground, hills 500 ft high and 2 nm apart, and a ridge 800 ft high whose 15° faces are steeper than any climb of the aircraft's. Over level ground the pitch holds within 0.01° (rms), 0.02° with the timer's jitter. Over the hills at 120 kt the clearance stays within 198–202 ft; at 200 kt the valleys are narrower than the 2000 ft/min descent and the ride allow the aircraft to follow, and it stays up to 500 ft over them (mean 325). The ridge leaves at least 194 ft at 120 to 200 kt, its edge flown at the climb's limit. The vertical acceleration stays within 0.26 g, the ride's 0.2 g with what the height loop adds, and no run flies up. The timer's jitter changes the least clearance by at most 1 ft. At 170 kt, with the nacelles at 20° and the lever near its stop, the climb lags its model and the ridge leaves 178 ft. The plan's 45 s are the time to clear 1000 ft of rise at the conversion's least climb (1449 ft/min at 150 kt) with the pull-up; higher or nearer terrain, or terrain coming into view round a turn, can be cleared only as the fly-up allows.

6 Corrections found during validation

The changes that the comparisons forced are listed here because each removed a physical or numerical inconsistency rather than tuning a coefficient.

1. *The sections' stall.* The sections first had C-81's NACA 0012 stall, the lift falling to a flat plate's within 6°. Its negative lift slope past the maximum took away the gimbal's aerodynamic damping, and at high thrust the gimbal diverged. The 64-series sections stall from the trailing edge and fall gently [12]; so do the model's now, the shape fitted to the hover test.
2. *The nacelles' mass.* Each nacelle's centre of gravity was placed on the wrong side of the conversion axis, which put the aircraft's 5 in forward.
3. *The governor.* Its error had the wrong sign: a rotor slower than its reference took more collective, so it slowed further. And its engagement was gated at an absolute rotor speed of 85–95 %, which turned it off in airplane mode at 84 %; it is gated relative to the reference now.
4. *The flaperons in slow airplane mode.* Retracted with the nacelles down, they left the wing unable to carry the aircraft at the corridor's lowest airplane-mode speeds; the schedule keeps 40° there.
5. *Test weights.* The tests' weight changes went into the cargo's point mass, which moved the centre of gravity with them; they are at the design centre of gravity now.
6. *The drive's torque limit* was applied as a power proportional to the rotor speed, so a start had almost no torque to turn the rotors up; it acts as a torque now, and the flight idle is the power the rotors need at flat pitch at their governed speed.
7. *The gear in the comparisons.* The harness first ran with the gear extended, JSBSim's default. Its drag stood in for drag the airframe lacks: the V-22's fitted drag (then 4.7 ft^2) met Boeing's polar and the airplane-mode speeds only with the gear down, and the XV-15's airplane-mode power looked close. All comparisons in forward flight are now gear up, the V-22's drag refitted, and the XV-15's shortfall reported (Section 5.2).
8. *The nacelle rate.* The nacelles' tilt rate, which moves the hubs and adds to the shafts' pitch rate in the flapping, was taken after the step had already set the new nacelle angle, so it was zero throughout. Taken from the step itself it changes no trim, and the pitch attitude through the conversion run (Section 5.7) by up to 1.6°.
9. *The handling runs' start.* The harness let the aircraft go one frame after setting the lever and centring the stick, before the controls' chain (the mixing, its actuators, the governor) had carried them through, and set the augmentation's integrators from that frame's values: the hover run began with a 2 % sag in rotor speed, drifted 3 kt and lost 40 ft that the aircraft does not. It now waits a whole period of the controls' channels (four frames) before letting go, and the hover holds its rotor speed and height from the start.
10. *The trim feed-forward in rearward flight.* The augmentation's trim feed-forward looked its stick up against the airspeed's magnitude, so that flying backwards it gave the forward stick of forward flight: in FlightGear the aircraft let go in the hover drifted aft, pitched nose down and diverged with the augmentation's integrator at its limit. It takes the

Table 11: Summary: ✓ within the test's scatter, ~ partly, × not reproduced. Fitted: the comparison checks the shape only.

XV-15 rotor: hover power, C_T/σ = 0.08–0.18 (stall fitted)	✓
XV-15 rotor: hover power at low thrust; collective gradient	~
XV-15 download against flaps (fitted)	✓
XV-15 power through the conversion (altitudes bracketed)	✓
XV-15 power in helicopter mode, 45–94 kt	×
XV-15 power in airplane mode, gear up (Bell's drag)	×
V-22 airframe polar (drag fitted)	✓
V-22 maximum level speed, sea level and 17 000 ft	✓
V-22 hover ceiling (sea level fitted)	✓
V-22 download at 67°	~
V-22 pitch attitude through the conversion	✓
V-22 wing lift share through the conversion	~

forward airspeed now (the hover's stick in rearward and sideward flight), carried past 90° of nacelle along its slope from 75 to 90.

11. *Yaw at low speed.* The augmentation only damped the yaw rate, through a washout: a steady yaw moment, a tailwind on the fins, turned the aircraft at will, and the pedals, opposed by the damper, were weak (a keyboard step turned it 0.6°). Below 50 kt the pedals now command a yaw rate, held by an integrator with the pedals' full travel.
12. *The elevator trim.* The original aircraft's controls file set FlightGear's elevator trim to 0.25 at the start, a YASim setting that the augmentation took as a quarter of the stick held forward: coupled, the flight director's pitch cue sat 5° above the aircraft. It starts at 0 now.

7 Assessment

Table 11 summarises. The model reproduces what defines the tiltrotor's performance: the proprotor's hover power at operating thrust, the wing's download against its flaps, the airframe's polar, and the V-22's speeds and hover ceilings, the last three after one fitted drag area and one fitted torque limit, both against Boeing's own charts. Through the conversion the V-22's pitch attitude follows its flight test. Its open items are the helicopter mode's power against speed, where the XV-15's test shows a shallower bucket than the blade elements and the wake give; the drag Bell's coefficients lack, which the V-22 build supplies by a fit and the XV-15 build leaves; the wing's share of the lift at intermediate nacelle angles; and the V-22 wing's download. None of the V-22's dynamic responses could be compared: no frequency responses or time histories of the aircraft are public, and the XV-15's, which are [23], are left for further work. Its augmentation is a plausible control law, not the V-22's.

8 Conclusions

- A blade-element model of the gimballed tiltrotor, both rotors blade by blade with their gimbals, pitch–flap coupling and finite-state wakes, the wing in their ring-vortex wakes, the interconnected drive and the conversion corridor, runs in plain JSBSim, generated from data. It is the flight model of the FlightGear V-22.
- Built first as the XV-15 from Bell's simulation data and NASA's tests, its rotor gives the full-scale rotor's hover power at operating thrust, and its download follows the flight test's with the flap.
- Carried to the V-22 with its published data, one fitted drag area and one fitted torque limit, it gives Boeing's airframe polar, maximum speeds within 15 kt and hover ceilings within 5 %, and the flight test's pitch attitude through the conversion.
- Bell's airframe coefficients miss part of the flying aircraft's drag; with the gear up the XV-15 build needs a quarter to a third less power than its flight test in airplane mode.
- The helicopter mode's power against speed, the wing's lift through the middle of the conversion and the V-22's download are documented, not tuned away.
- Its autopilot's terrain following plans a reference path over the terrain ahead and holds the aircraft to its measured response: over hills and a ridge steeper than its climb it keeps within 4 % of a 200 ft clearance at 120 to 200 kt, without porpoising over level ground.

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